$\sqrt{\quad 3R}$

11 *Ans:* ***D***

$F_c = ma_c$

$mg = mr\omega^2$

When satellite is orbiting near the Moon's surface,

$$m\frac{g}{6} = m\left(\frac{R}{4}\right)\left(\frac{2\pi}{T_M}\right)^2 \qquad \text{--- (1)}$$

When satellite is orbiting near the Earth's surface

$$mg = mR\left(\frac{2\pi}{T}\right)^2 \qquad \text{--- (2)}$$

$$\frac{(1)}{(2)}: \frac{m\frac{g}{6}}{mg} = \frac{m\left(\frac{R}{4}\right)\left(\frac{2\pi}{T_M}\right)^2}{mR\left(\frac{2\pi}{T}\right)^2}$$

$$\left(\frac{T_M}{T}\right)^2 = \frac{6}{4}$$

$$T_M = \sqrt{\frac{3}{2}}T$$

12 *Ans:* ***A***